

Variables for SDM

- Characteristics and source of environmental data
- Processing environmental data for ENM/SDM

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Correlative approach:



openModeller

<http://openmodeller.sourceforge.net>

Environmental data/Raster data:

➤ Why call it layer?

Demonstrating → ArcGIS/ArcPro, QGIS

➤ Format:

.asc

.tif

.bil,

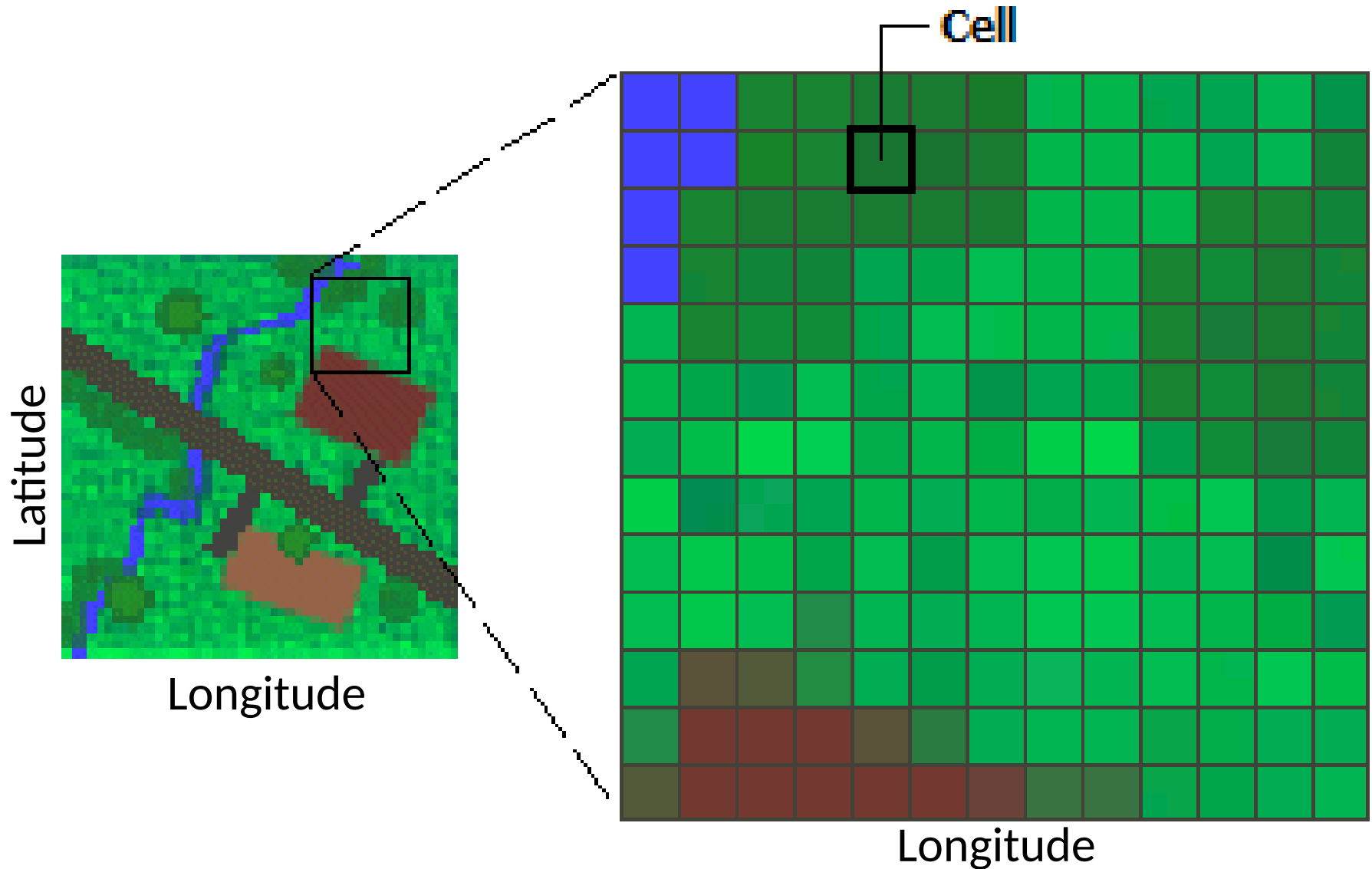
.rst,

.txt,

.img,

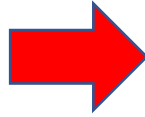
.netCDF,

What is a raster data looks like?



Data structures and object types in R

Raster data



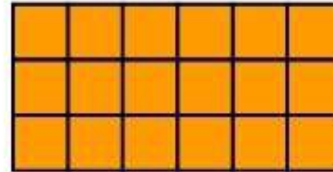
Scalar



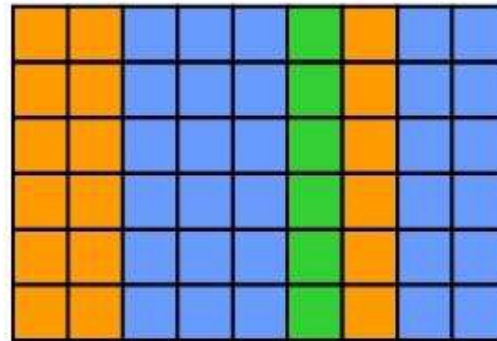
Vector



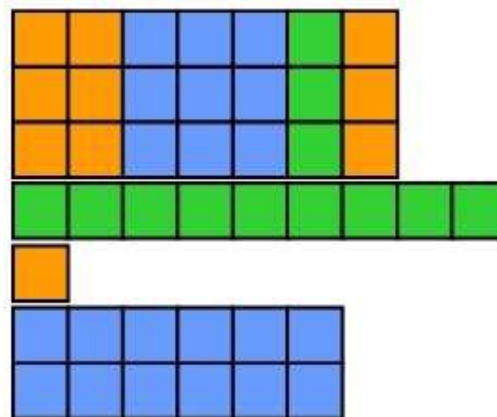
Matrix



Data Frame



List



Data type

-  Numerical
-  Character
-  Logical

File

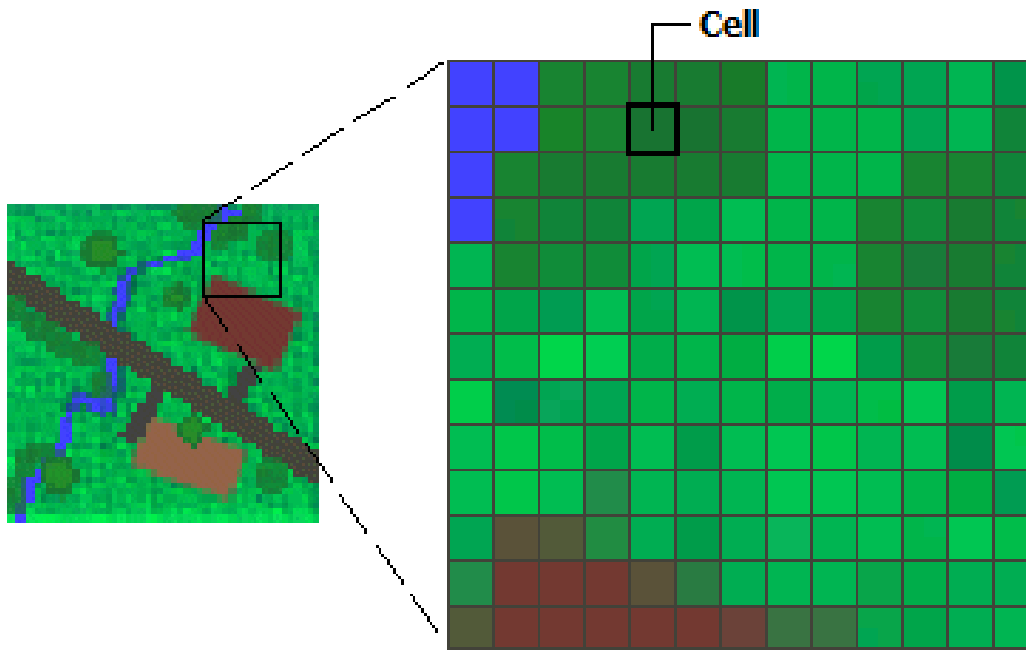
Edit

View

[illegible]

General characteristics of raster data

```
> r1<-raster("Washington.tif")
> r1
class       : RasterLayer
dimensions  : 414, 938, 388332  (nrow, ncol, ncell)
resolution  : 0.008333333, 0.008333333  (x, y)
extent      : -124.7333, -116.9167, 45.55, 49  (xmin, xmax, ymin, ymax)
crs         : +proj=longlat +datum=WGS84 +no_defs
source      : Washington.tif
names       : layer
values      : 0.00189019, 0.817575  (min, max)
```



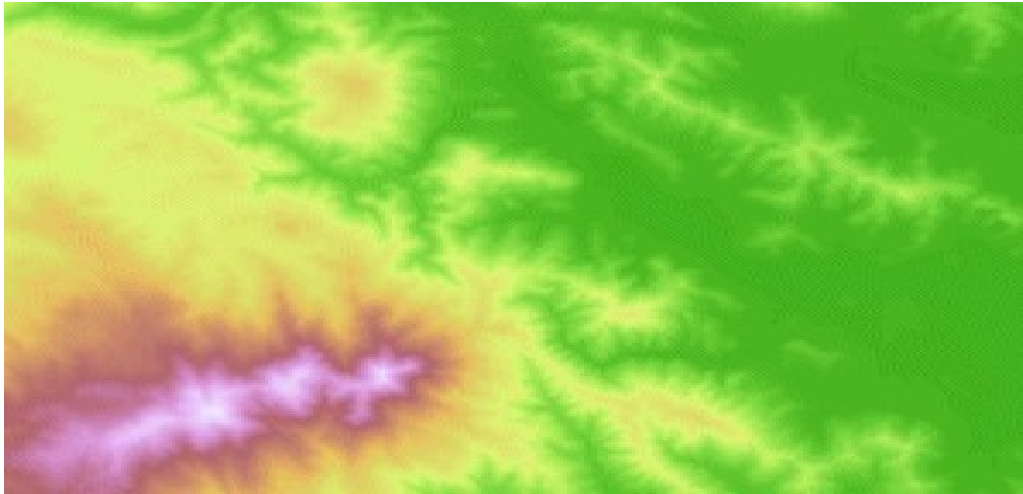
Unifying raster layers- why?

- Many times, we are using environmental variables from different sources;

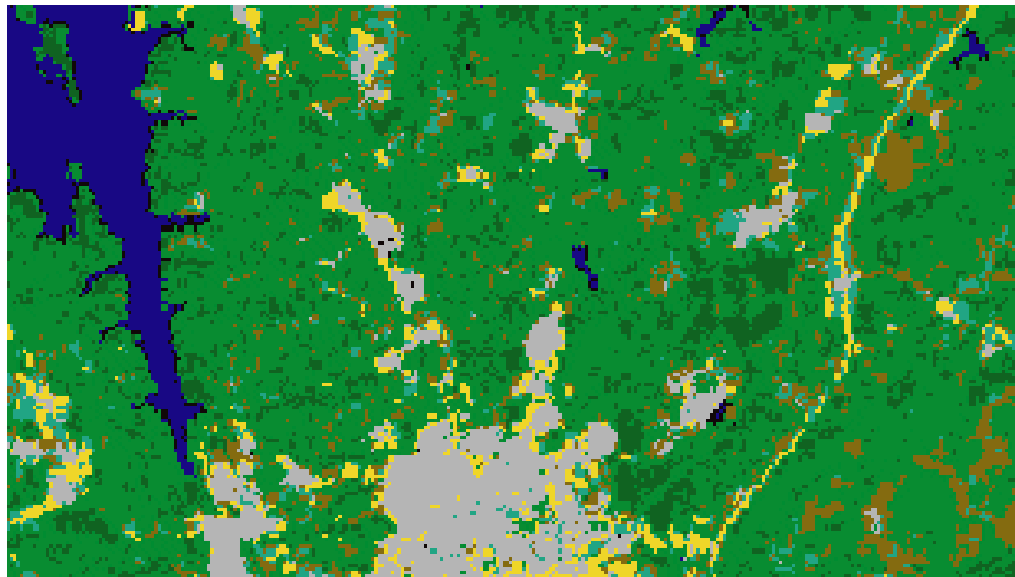
e.g., climate, topographic, landcover, et al., these variables are provided by different originations, they use their unique way to deposit/organize the data...

- These variables/data have different extent and resolution.., you need to unify them before fitting the model;

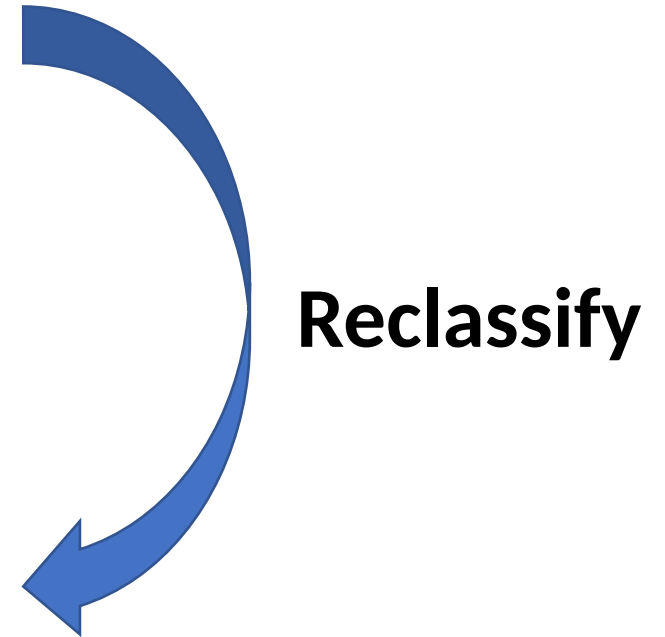
Raster as surface maps---continuous data



Raster as thematic maps--- categorized data

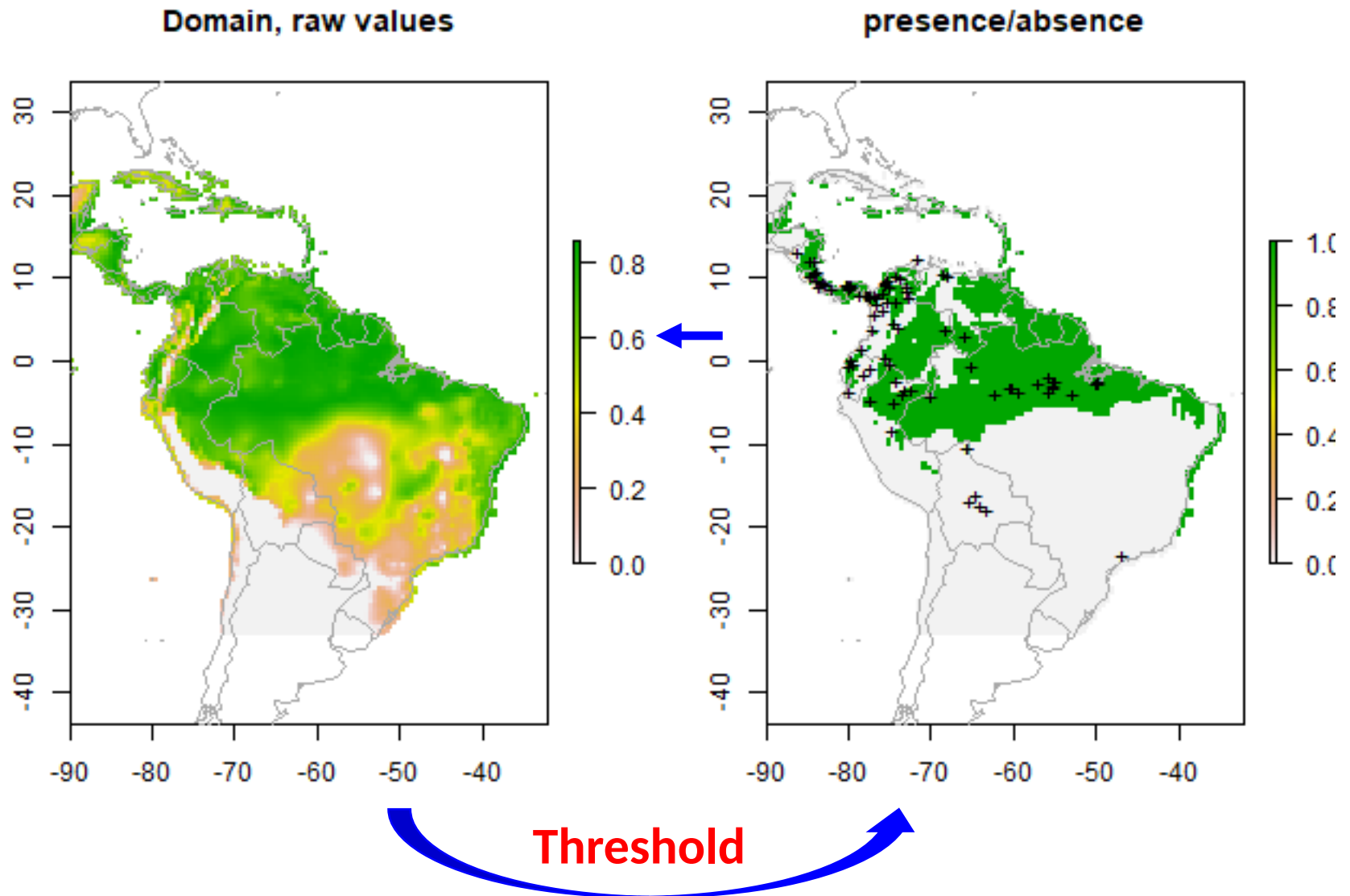


- | | |
|--|---|
|  Agriculture |  Grass |
|  Bare ground |  Pine |
|  Water |  Shadow |
|  Deciduous |  Urban/Developed |
|  Deciduous/
Pine mixed | |

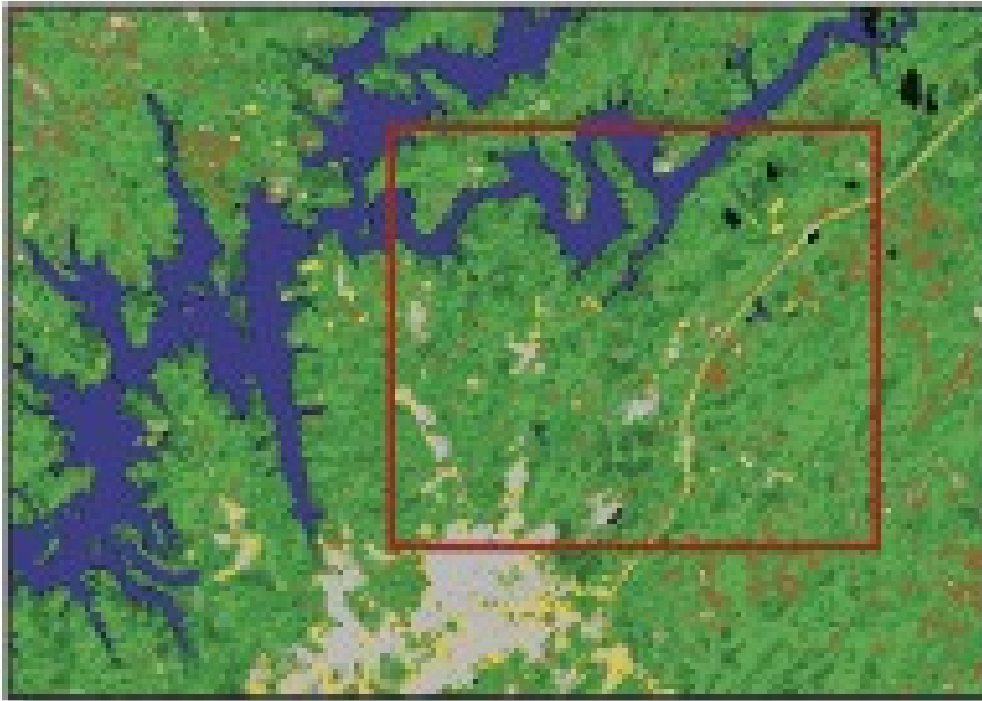


Reclassify-

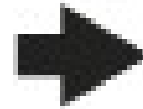
suitability prediction → binary prediction



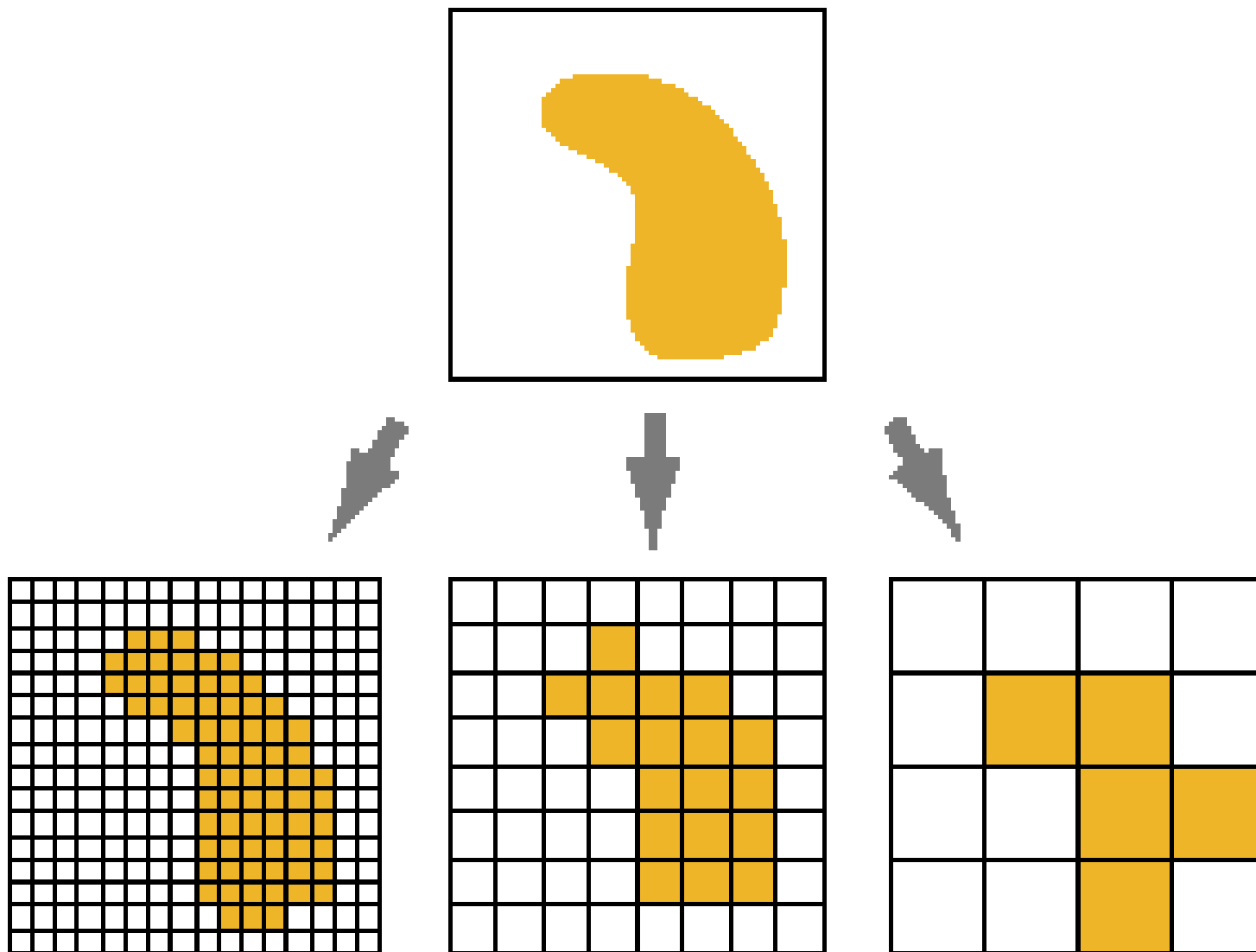
Change extent-- clip



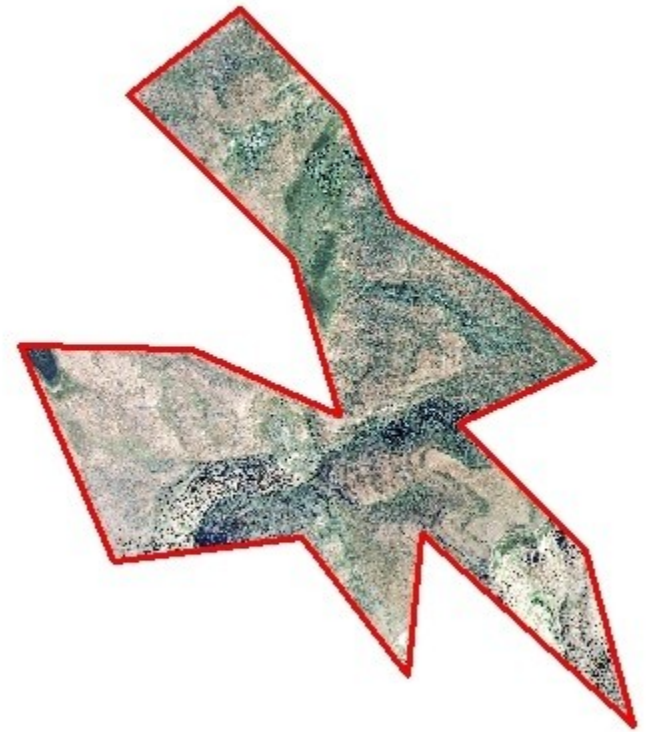
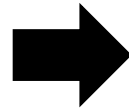
Clip

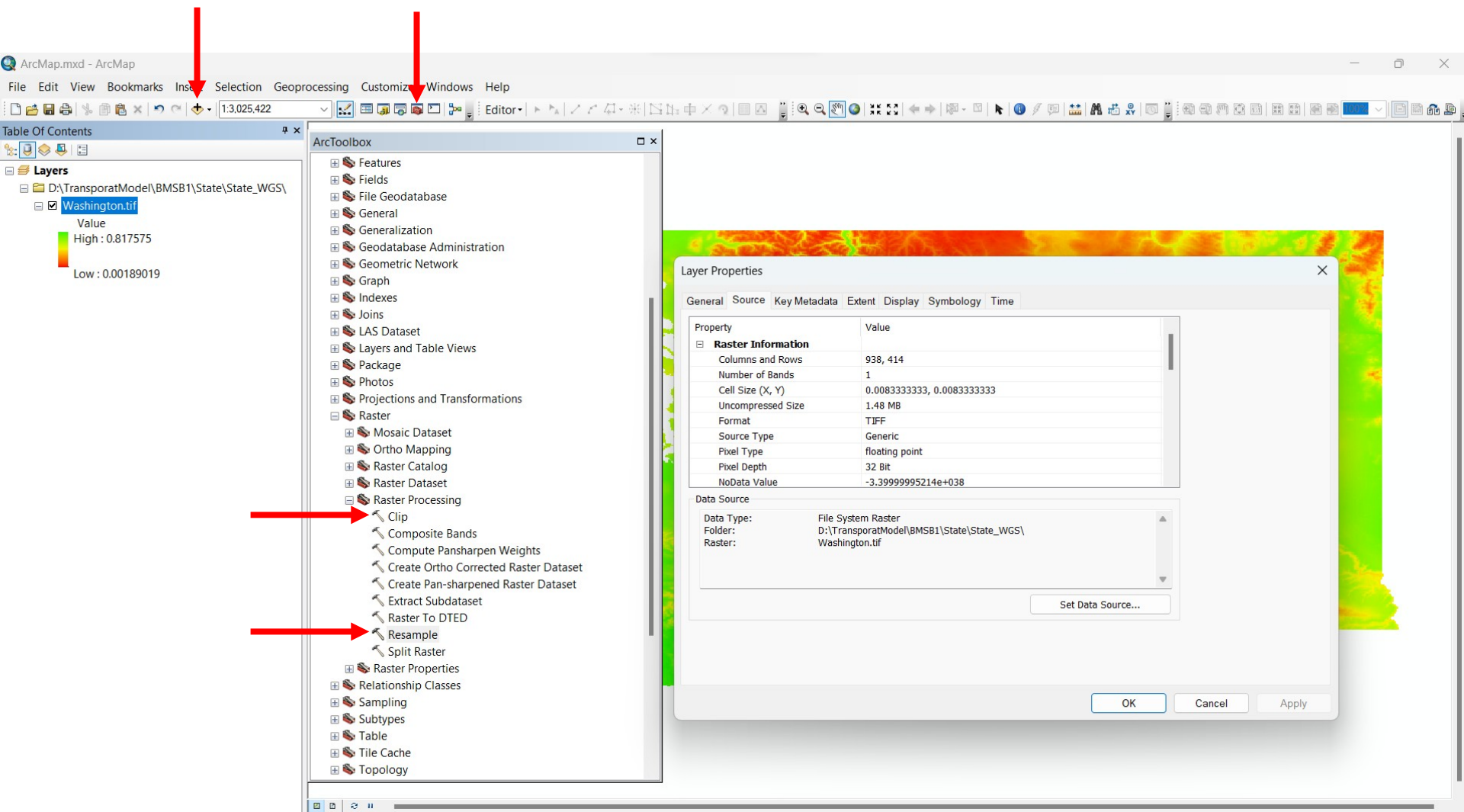


Change resolution--resample



Extract by mask- raster and polygon

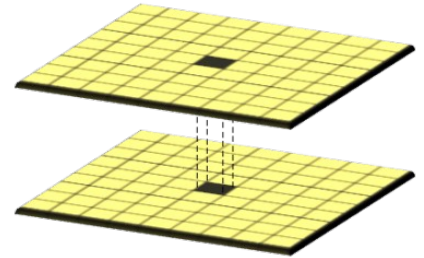




Classical map algebra

-local analysis

- Local: cell-by-cell basis



1	4	5
5	3	2
2	5	2

 +

5	1	3
1	2	1
1	4	2

 =

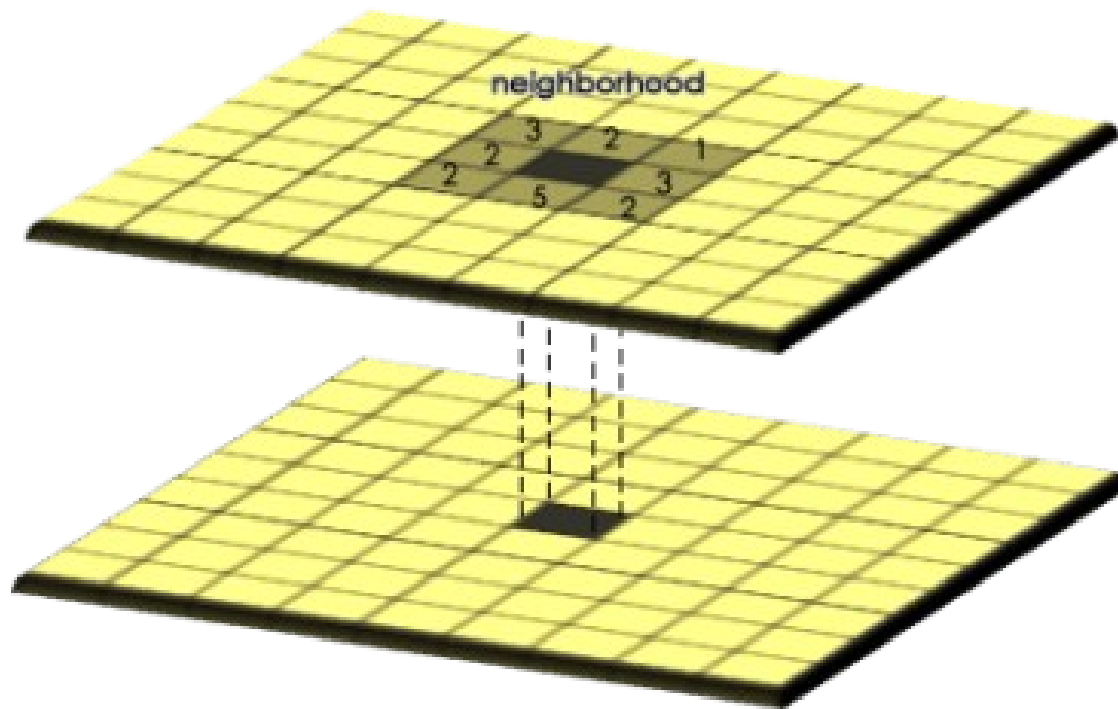
6	5	8
6	5	3
3	9	4

Classical map algebra

-Focal analysis

- **Focal:**

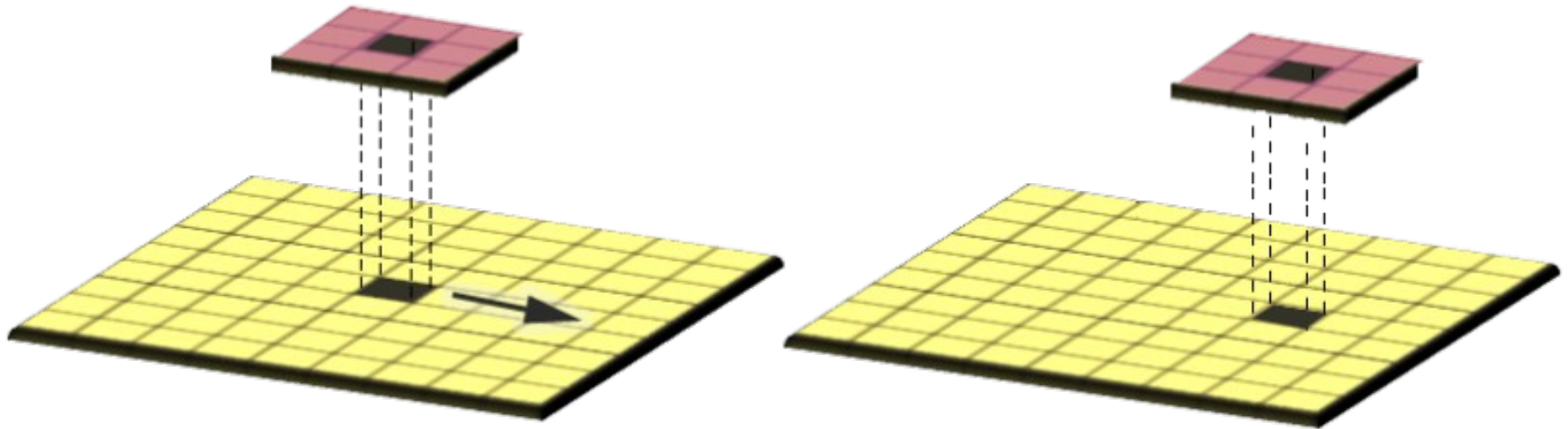
compute an output value of each cell using neighborhood values.



Map algebra/Raster Math- **Focal**

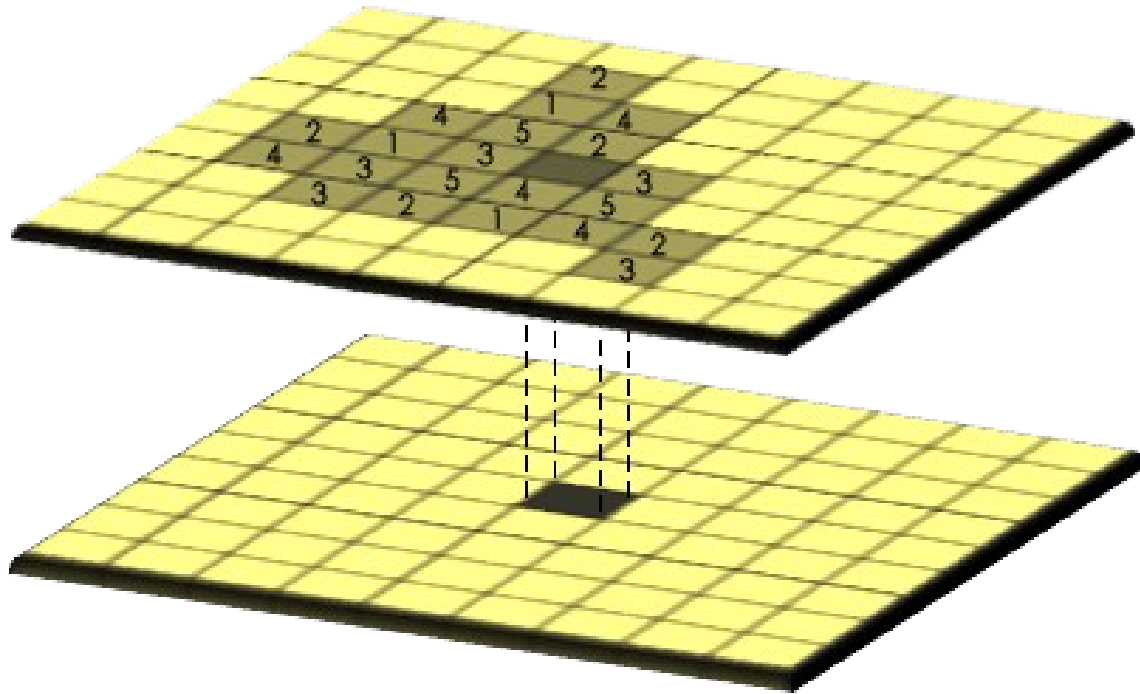
- **Focal:**

compute an output value of each cell using neighborhood values.



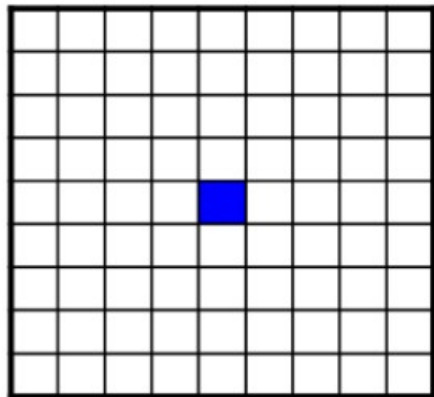
Map algebra-Zonal Operation

Zonal operations apply a math function to a group of cells within a specified zone.

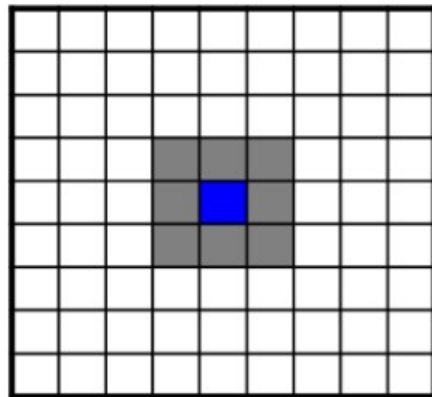


Map Algebra Operations

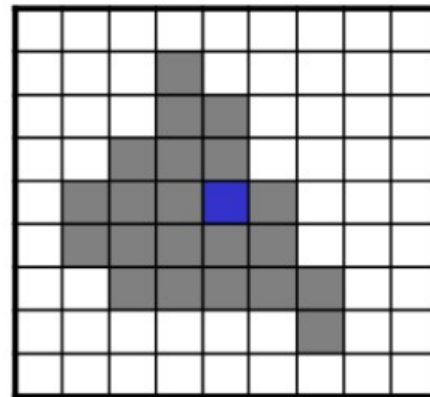
- Tomlin (1990) defines and organizes operations as *local*, *focal*, *zonal*, and *global* according to the *spatial scope* of the operations
 - *Geographic Information System and Cartographic Modeling*, Englewood Cliffs: Prentice Hall, 1990.



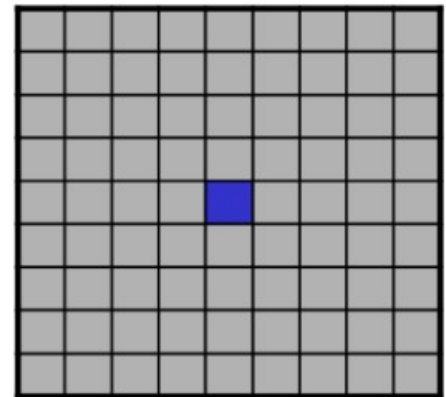
Local



Focal

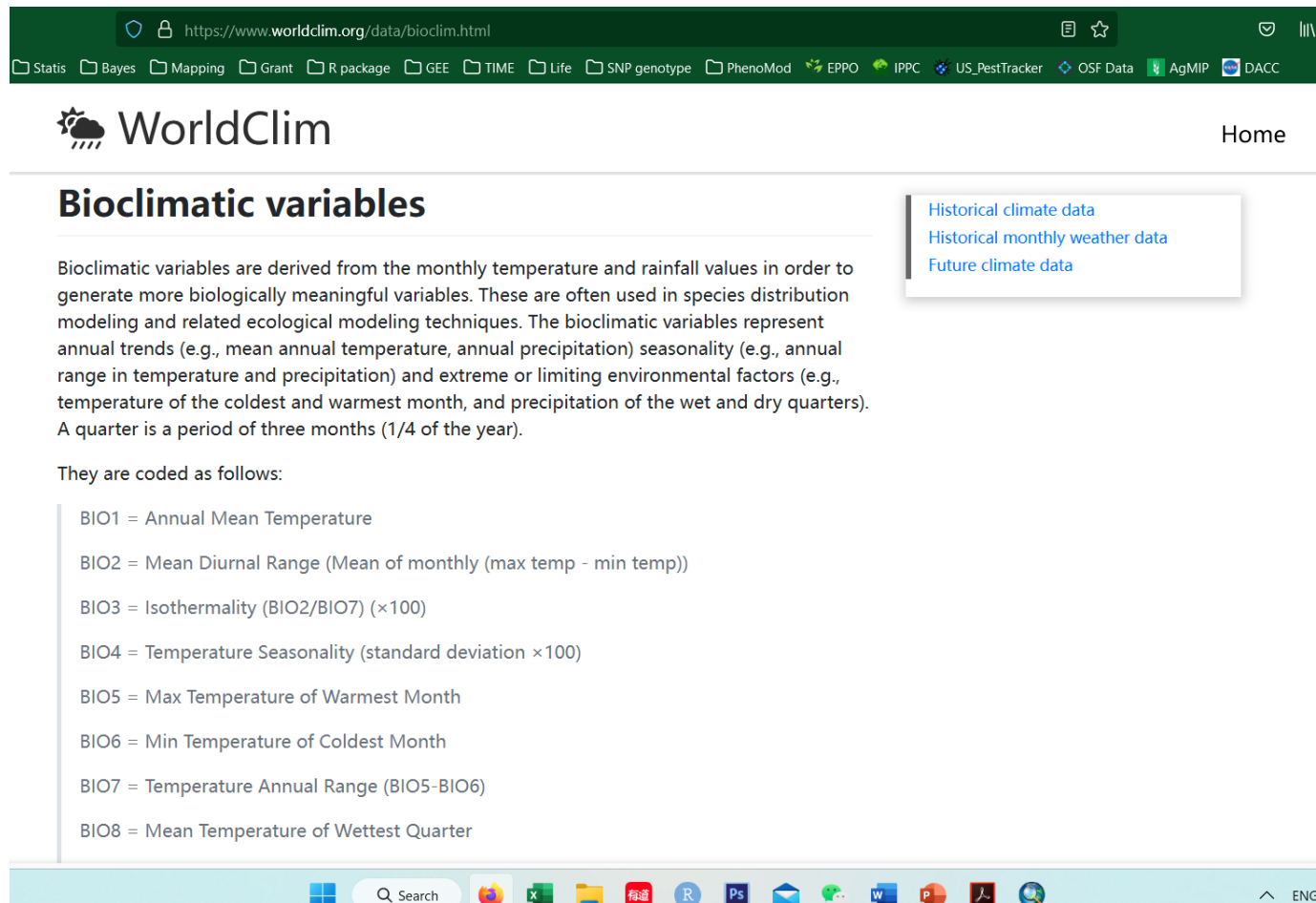


Zonal



Global

Source of environmental/raster data for model: WorldClim-bioclimate data



The screenshot shows the WorldClim website interface. The browser address bar displays <https://www.worldclim.org/data/bioclimate.html>. The website header includes the WorldClim logo and a 'Home' link. The main section is titled 'Bioclimatic variables' and contains a paragraph explaining that these variables are derived from monthly temperature and rainfall values for use in species distribution modeling. To the right of this text is a dropdown menu with three options: 'Historical climate data', 'Historical monthly weather data', and 'Future climate data'. Below the paragraph, a list of eight bioclimatic variables (BIO1 to BIO8) is provided, each with its full name and calculation formula. The website footer shows a Windows taskbar with various application icons and the system language set to 'ENG'.

WorldClim Home

Bioclimatic variables

Bioclimatic variables are derived from the monthly temperature and rainfall values in order to generate more biologically meaningful variables. These are often used in species distribution modeling and related ecological modeling techniques. The bioclimatic variables represent annual trends (e.g., mean annual temperature, annual precipitation) seasonality (e.g., annual range in temperature and precipitation) and extreme or limiting environmental factors (e.g., temperature of the coldest and warmest month, and precipitation of the wet and dry quarters). A quarter is a period of three months (1/4 of the year).

They are coded as follows:

- BIO1 = Annual Mean Temperature
- BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))
- BIO3 = Isothermality ($BIO2/BIO7$) ($\times 100$)
- BIO4 = Temperature Seasonality (standard deviation $\times 100$)
- BIO5 = Max Temperature of Warmest Month
- BIO6 = Min Temperature of Coldest Month
- BIO7 = Temperature Annual Range ($BIO5 - BIO6$)
- BIO8 = Mean Temperature of Wettest Quarter

<https://www.worldclim.org>

BIO1 = Annual Mean Temperature ↑

BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))

BIO3 = Isothermality (BIO2/BIO7) ($\times 100$)

BIO4 = Temperature Seasonality (standard deviation $\times 100$)

BIO5 = Max Temperature of Warmest Month

BIO6 = Min Temperature of Coldest Month

BIO7 = Temperature Annual Range (BIO5-BIO6)

BIO8 = Mean Temperature of Wettest Quarter

BIO9 = Mean Temperature of Driest Quarter

BIO10 = Mean Temperature of Warmest Quarter

BIO11 = Mean Temperature of Coldest Quarter

BIO12 = Annual Precipitation ↑

BIO13 = Precipitation of Wettest Month

BIO14 = Precipitation of Driest Month

BIO15 = Precipitation Seasonality (Coefficient of Variation)

BIO16 = Precipitation of Wettest Quarter

BIO17 = Precipitation of Driest Quarter

BIO18 = Precipitation of Warmest Quarter

BIO19 = Precipitation of Coldest Quarter

19 Bioclimatic variables

**Temperature and
precipitation combination**

**Temperature and
precipitation combination**

The data is available at the four spatial resolutions, between 30 seconds (~1 km²) to 10 minutes (~340 km²). Each download is a "zip" file containing 12 GeoTiff (.tif) files, one for each month of the year (January is 1; December is 12).

variable	10 minutes	5 minutes	2.5 minutes	30 seconds
minimum temperature (°C)	tmin 10m	tmin 5m	tmin 2.5m	tmin 30s
maximum temperature (°C)	tmax 10m	tmax 5m	tmax 2.5m	tmax 30s
average temperature (°C)	tavg 10m	tavg 5m	tavg 2.5m	tavg 30s
precipitation (mm)	prec 10m	prec 5m	prec 2.5m	prec 30s
solar radiation (kJ m ⁻² day ⁻¹)	srad 10m	srad 5m	srad 2.5m	srad 30s
wind speed (m s ⁻¹)	wind 10m	wind 5m	wind 2.5m	wind 30s
water vapor pressure (kPa)	vapr 10m	vapr 5m	vapr 2.5m	vapr 30s

Below you can download the standard (19) WorldClim [Bioclimatic variables](#) for WorldClim version 2. They are the average for the years 1970-2000. Each download is a "zip" file containing 19 GeoTiff (.tif) files, one for each month of the [variables](#).

variable	10 minutes	5 minutes	2.5 minutes	30 seconds
Bioclimatic variables	bio 10m	bio 5m	bio 2.5m	bio 30s

Future climate, 2.5 minutes spatial resolution

See the [main page](#) for background information and for other spatial resolutions. Downscaled monthly future climate data from [CMIP6](#) is available in GeoTiff files for nine GCMs and four SSPs for the following time periods: [2021-2040](#), [2041-2060](#), [2061-2080](#), and [2081-2100](#). The climate variables available are tn - monthly average minimum temperature (°C), tx - monthly average maximum temperature (°C), pr - monthly total precipitation (mm), and bc - [bioclimatic](#) variables.

2021-2040

GCM	ssp126	ssp245	ssp370	ssp585
ACCESS-CM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
ACCESS-ESM1-5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
BCC-CSM2-MR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5-CanOE	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CMCC-ESM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1-HR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc

2061-2080

GCM	ssp126	ssp245	ssp370	ssp585
ACCESS-CM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
ACCESS-ESM1-5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
BCC-CSM2-MR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5-CanOE	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CMCC-ESM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1-HR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-ESM2-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg-LR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
FIO-ESM-2-0	tn, tx, pr, bc	tn, tx, pr, bc		tn, tx, pr, bc

2041-2060

GCM	ssp126	ssp245	ssp370	ssp585
ACCESS-CM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
ACCESS-ESM1-5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
BCC-CSM2-MR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CanESM5-CanOE	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CMCC-ESM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1-HR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-ESM2-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg-LR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
FIO-ESM-2-0	tn, tx, pr, bc	tn, tx, pr, bc		tn, tx, pr, bc

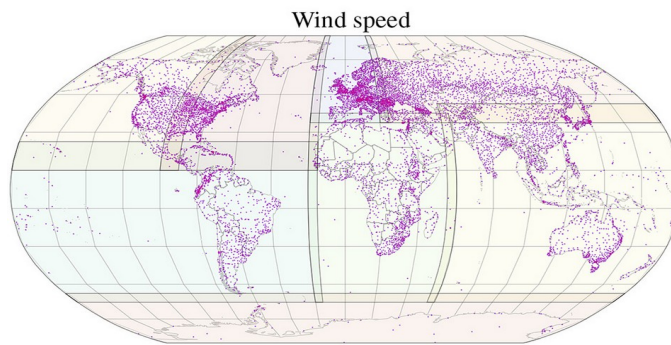
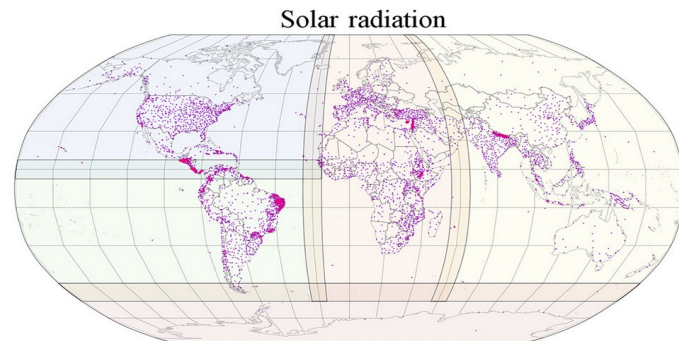
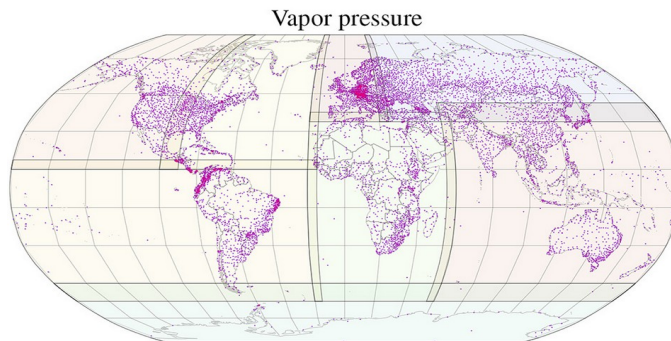
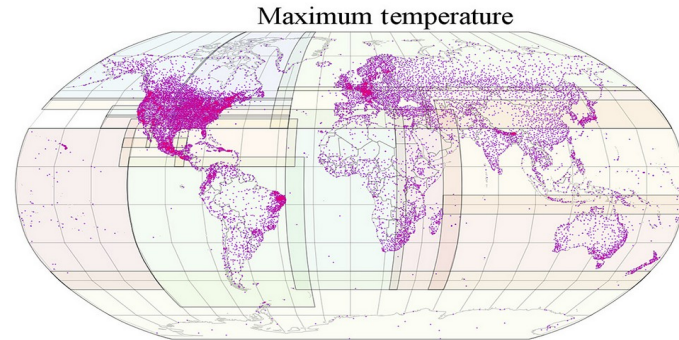
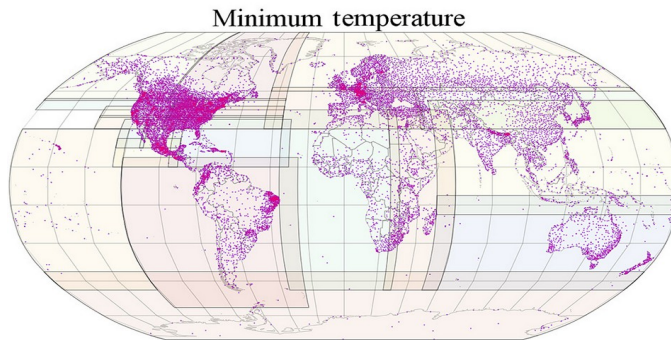
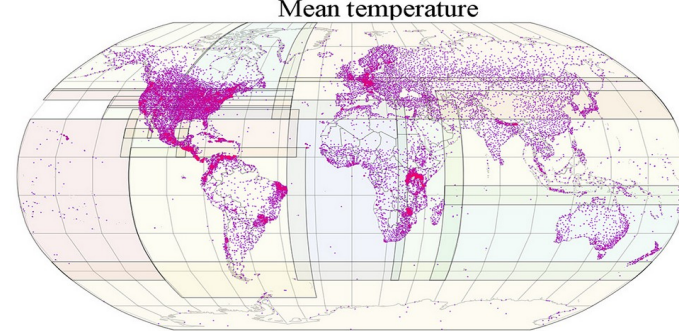
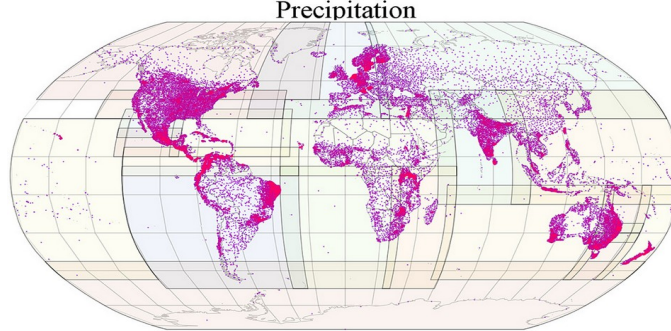
2081-2100

GCM	ssp126	ssp245	ssp370	ssp585
ACCESS-CM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
ACCESS-ESM1-5	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
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CanESM5-CanOE	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CMCC-ESM2	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-CM6-1-HR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
CNRM-ESM2-1	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
EC-Earth3-Veg-LR	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc	tn, tx, pr, bc
FIO-ESM-2-0	tn, tx, pr, bc	tn, tx, pr, bc		tn, tx, pr, bc

How these data
generated?
From weather station
by interpolation
method.

WorldClim 2

**There is spatial
uncertainty...**



**Global spatial distribution
of weather stations**

Home

Downloads

Daily timeseries

Monthly timeseries

Bioclimate / Köppen-Geiger

Future

Paleo Climate

Climate diagrams



Climatologies at high resolution for the earth's land surface areas

Bioclim

CHELSA Bioclim

Bioclimatic variables are derived variables from the monthly mean, max, mean temperature, and mean precipitation values. They are developed for species distribution modeling and related ecological applications. They represent annual trends (e.g., mean annual temperature, annual precipitation) seasonality (e.g., annual range in temperature and precipitation) and extreme or limiting environmental factors (e.g., temperature of the coldest and warmest month, and precipitation of the wet and dry quarters). A quarter is defined as a

AdaptWest - A Climate Adaptation Conservation Planning Database for North America

Search by keyword or location



powered by



Get Started

Browse Data

Create

Community

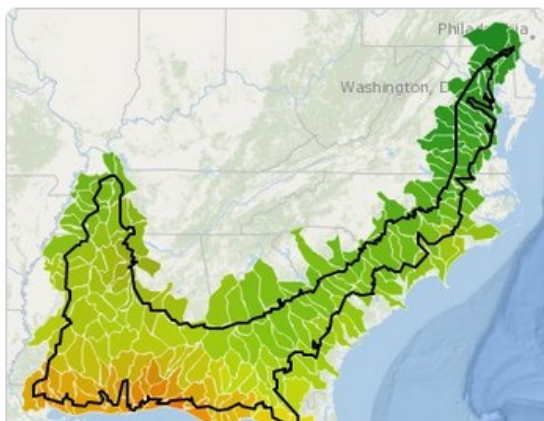
Workspace

DATA BASIN

Welcome to the AdaptWest Climate Adaptation Data Portal!

AdaptWest is a spatial database designed to help land management agencies and other organizations implement strategies that promote resilience, protect biodiversity, and conserve and enhance the adaptation potential of natural systems in the face of a changing climate.

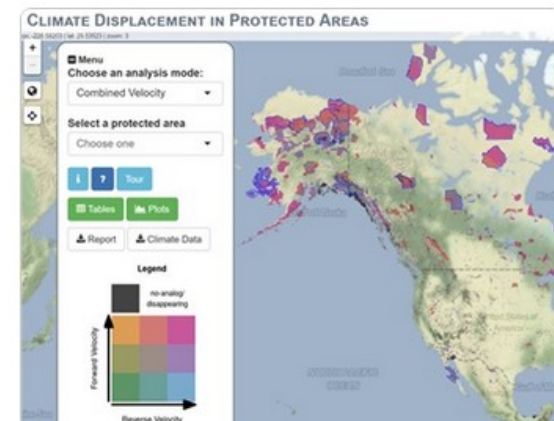
Watershed Climate Data Explorer



Climate Resilience Data Explorer



Climate Displacement Explorer



Bioclimatic variables in AdaptWest Climate center

Bioclimatic variables:

MAT: mean annual temperature (°C)
MWMT: mean temperature of the warmest month (°C)
MCMT: mean temperature of the coldest month (°C)
TD: difference between MCMT and MWMT, as a measure of continentality (°C)
MAP: mean annual precipitation (mm)
MSP: mean summer (May to Sep) precipitation (mm)
AHM: annual heat moisture index, calculated as $(MAT+10)/(MAP/1000)$
SHM: summer heat moisture index, calculated as $MWMT/(MSP/1000)$
DD_0: degree-days below 0°C (chilling degree days)
DD5: degree-days above 5°C (growing degree days)
DD_18: degree-days below 18°C
DD18: degree-days above 18°C
NFFD: the number of frost-free days
FFP: frost-free period
bFFP: the julian date on which the frost-free period begins
eFFP: the julian date on which the frost-free period ends
PAS: precipitation as snow (mm)
EMT: extreme minimum temperature over 30 years
EXT: extreme maximum temperature over 30 years
Eref: Hargreave's reference evaporation
CMD: Hargreave's climatic moisture index
MAR: mean annual solar radiation (MJ m⁻² d⁻¹) (excludes areas south of US and some high-latitude areas)
RH: mean annual relative humidity (%)
CMI: Hogg's climate moisture index (mm)
DD1040: (10<DD<40) degree-days above 10°C and below 40°C
Tave_wt: winter (December to February) mean temperature (°C)
Tave_sp: spring (March to May) mean temperature (°C)
Tave_sm: summer (June to August) mean temperature (°C)
Tave_at: autumn (September to November) mean temperature (°C)
PPT_wt: winter (December to February) precipitation (mm)
PPT_sp: spring (March to May) precipitation (mm)
PPT_sm: summer (June to August) precipitation (mm)
PPT_at: autumn (September to November) precipitation (mm)

Bio-ORACLE

Marine data layers for ecological modelling

The latest release of the Bio-Oracle dataset is version 2.2. [Click here](#) for more information.



Extensive surface and benthic dataset

Bio-ORACLE is a set of GIS rasters providing geophysical, biotic and environmental data for surface and benthic marine realms.



Uniform and worldwide

The data are available for global-scale applications at a spatial resolution of 5 arcmin (approximately 9.2 km at the equator).



Forecasting and transferability

The most recent **Representative Concentration Pathways** are provided in order to model the ecological implications of future changes.

What is its potential?

Linking biodiversity occurrence data to the physical and biotic environment provides a framework to formulate hypotheses about the ecological processes governing spatial and temporal patterns in biodiversity, which can be useful for marine ecosystem management and conservation.

Bio-ORACLE offers a user-friendly solution to accomplish this task by providing 19 global geophysical, biotic and climate layers at a common spatial resolution (5 arcmin) and a uniform landmask.

Satellite data

Sources of satellite data (free)

1. MODIS (Moderate Resolution Imaging Spectroradiometer; 36 bands) global products

- Vegetation Indices (NDVI, EVI): 16-day or monthly; 2000-present; 250 m, 500 m, 1km
- Leaf Area Index and Fractional Photosynthetically Active Radiation: 4-day and 8-day; 2002-present; 500 m
- Gross Primary Productivity and Net Primary Productivity: 8-day; 2002-present; 500 m
- Evapotranspiration: 8-day and yearly; 2002-present; 500 m
- Land cover: yearly; 2001-2012; 500 m and 5,6 km
- Vegetation continuous fields (% tree cover; % non tree vegetation; % non vegetated): yearly; 2000-2013; 250 m

Satellite data

Sources of satellite data (free)

2. Global land cover based on 5 satellite missions: **yearly; 1992 to 2015; 300 m** <http://maps.elie.ucl.ac.be/CCI/viewer/download.php>
3. Global nighttime lights time series, 4 satellites: **yearly, 1992-2013; 1 km** <https://ngdc.noaa.gov/eog/dmsp/downloadV4composites.html>
4. Global Human Footprint: **average (1995-2004), 1 km** <http://sedac.ciesin.columbia.edu/data/set/wildareas-v2-human-footprint-geographic>
5. Elevation from SRTM: **90 m global (30 m US)** <https://lpdaac.usgs.gov>

Paleoclimate data---reconstruction historical distribution

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Downloads

Welcome to PaleoClim.org your source of free, high-resolution paleoclimate data for use in biological modeling and GIS.

Below you can download the standard [Bioclimatic variables](#) for different paleoclimate simulations. Each download is a "zip" file containing up to 19 GeoTiff (.tif) files, one for each of the bioclimatic variables.

The data are available at 3 different spatial resolutions; from 2.5 arc-minutes (~5 km), 5 arc-minutes (~10 km), and 10 arc-minutes (~20 km). The CHELSA data are also available in 30 arc-seconds (~1 km).



How to use these database?

- Go through database official website, read instructions/practice tutorial materials carefully;
- What are these data? check data structure and frame;
- What is the resolution, extent, coordinate system et al.?
- Are there specific software that used to download their data? they will recommend ways to process their data..

Ecological Niche Model

How many variables should be used?

- Too many variables

variable might be highly correlated...

- Too less variables

Might not enough to characterize species ecological requirement and potential distribution...

5-8
variables
would be
enough

Short communication

Sensitivity of distributional prediction algorithms to geographic data completeness

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Received 6 March 1998; accepted 8 December 1998

Abstract

The sensitivity of one algorithm for prediction of geographic distributions of species from point data to depth of geographic information was tested for three species of North American birds. Test species were chosen to represent three distinct distributional patterns—western North America (Pygmy Nuthatch *Sitta pygmaea*), eastern North America (Barred Owl *Strix varia*), and the Great Plains in the central part of the continent (Lark Bunting *Calamospiza melanocorys*). Distributional predictions were made using the expert-system algorithm Genetic Algorithm for Role-set Prediction (GARP). Depth of geographic information was manipulated by rarifying the number of coverages on which predictions were based, from the full complement of eight down to one, using a combination of jackknifing and bootstrapping. In all three species, five of the eight coverages were necessary to arrive at the asymptotic maximum predictive efficiency and to avoid broad variance in resulting predictive efficiencies. Annual mean temperature was a critical variable, in some cases more important than inclusion of additional coverages, to producing accurate distributional predictions. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Distributional prediction; Geographic coverages; Sensitivity; GARP; Biodiversity

How to select environmental variables?

- **Ecological relevance:**

How to rank variable importance?

expert opinion,

Many statistical approach,...

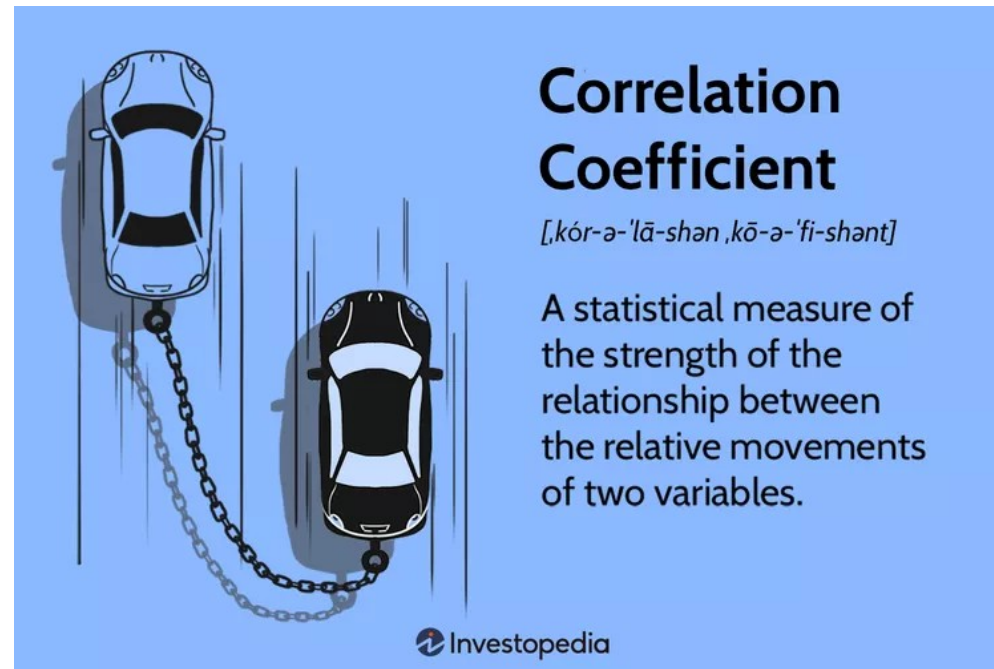
- **Correlation: Pearson < 0.7**

- **Resolution: cell size--resample**

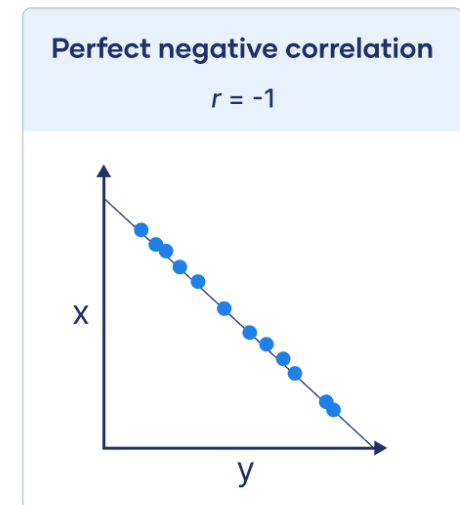
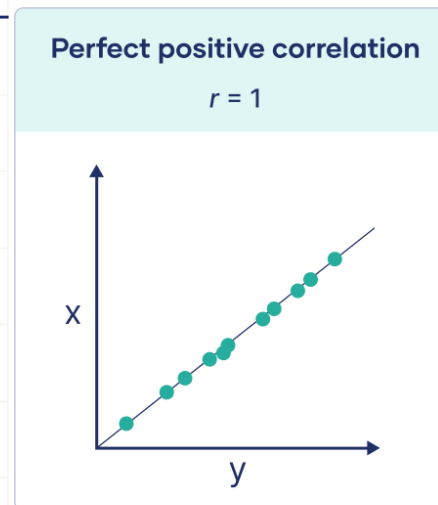
- **Extent: clip/crop, extract by mask**

Pearson correlation

- The **Pearson correlation coefficient (r)** is the most common way of measuring a linear correlation.
- It is a number between -1 and 1 that measures the strength and direction of the relationship between two variables.
- **Pearson < 0.7 is acceptable in our model!**



Pearson correlation coefficient (r) value	Strength	Direction
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and $-.3$	Weak	Negative
Between $-.3$ and $-.5$	Moderate	Negative
Less than $-.5$	Strong	Negative



ORIGINAL RESEARCH

Collinearity in ecological niche modeling: Confusions and challenges

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Funding information

The University of Arizona Office of Research, Discovery, and Innovation; Oklahoma State University, Grant/Award Number: NSF-OCI 1126330 University of Tennessee's Open Publishing Support Fund

Abstract

Ecological niche models are widely used in ecology and biogeography. Maxent is one of the most frequently used niche modeling tools, and many studies have aimed to optimize its performance. However, scholars have conflicting views on the treatment of predictor collinearity in Maxent modeling. Despite this lack of consensus, quantitative examinations of the effects of collinearity on Maxent modeling, especially in model transfer scenarios, are lacking. To address this knowledge gap, here we quantify the effects of collinearity under different scenarios of Maxent model training and projection. We separately examine the effects of predictor collinearity, collinearity shifts between training and testing data, and environmental novelty on model performance. We demonstrate that excluding highly correlated predictor variables does not significantly influence model performance. However, we find that collinearity shift and environmental novelty have significant negative effects on the performance of model transfer. We thus conclude that (a) Maxent is robust to predictor collinearity in model training; (b) the strategy of excluding highly correlated variables has little impact because Maxent accounts for redundant variables; and (c) collinearity shift and environmental novelty can negatively affect Maxent model transferability. We therefore recommend to quantify and report collinearity shift and environmental novelty to better infer model accuracy when models are spatially and/or temporally transferred.

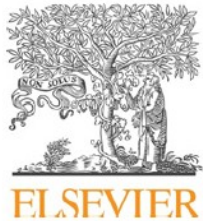
KEYWORDS

bioclim, collinearity shift, ecological niche, mammal, model transfer, predictor selection, species distribution model

Correlation is not an issue in
Maxent “calibrating”
prediction;
Feng et al. 2019 Eco & Evo

Why can't every model calibrate on global extent?

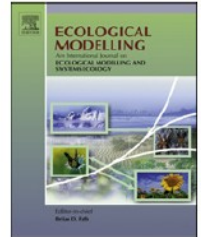
Ecological Modelling 222 (2011) 1810–1819



Contents lists available at ScienceDirect

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel



The crucial role of the accessible area in ecological niche modeling and species distribution modeling

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ARTICLE INFO

Article history:

Received 3 July 2010

Received in revised form 4 December 2010

Accepted 13 February 2011

Available online 7 April 2011

Keywords:

Accessible area

Area of distribution

Ecological niche

Geographic extent

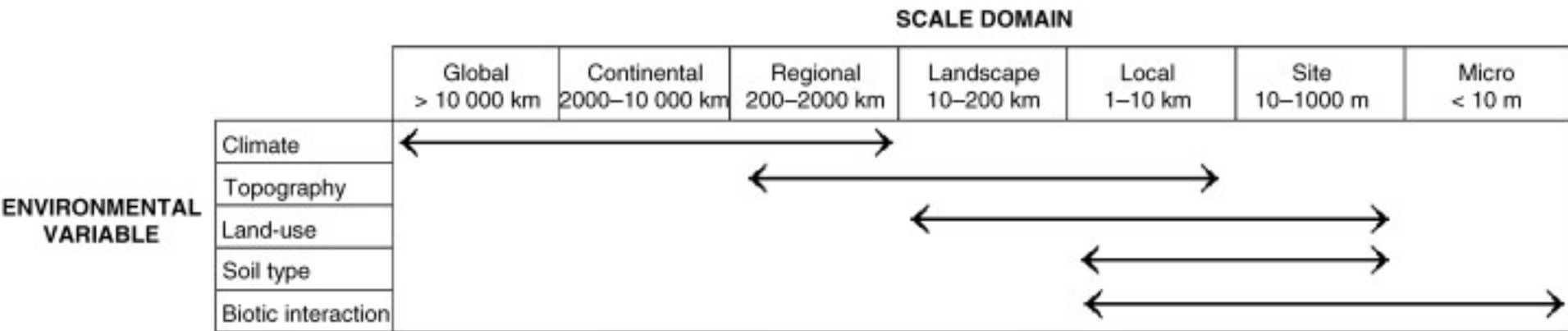
Pleistocene

ABSTRACT

Using known occurrences of species and correlational modeling approaches has become a common paradigm in broad-scale ecology and biogeography, yet important aspects of the methodology remain little-explored in terms of conceptual basis. Here, we explore the conceptual and empirical reasons behind choice of extent of study area in such analyses, and offer practical, but conceptually justified, reasoning for such decisions. We assert that the area that has been accessible to the species of interest over relevant time periods represents the ideal area for model development, testing, and comparison.

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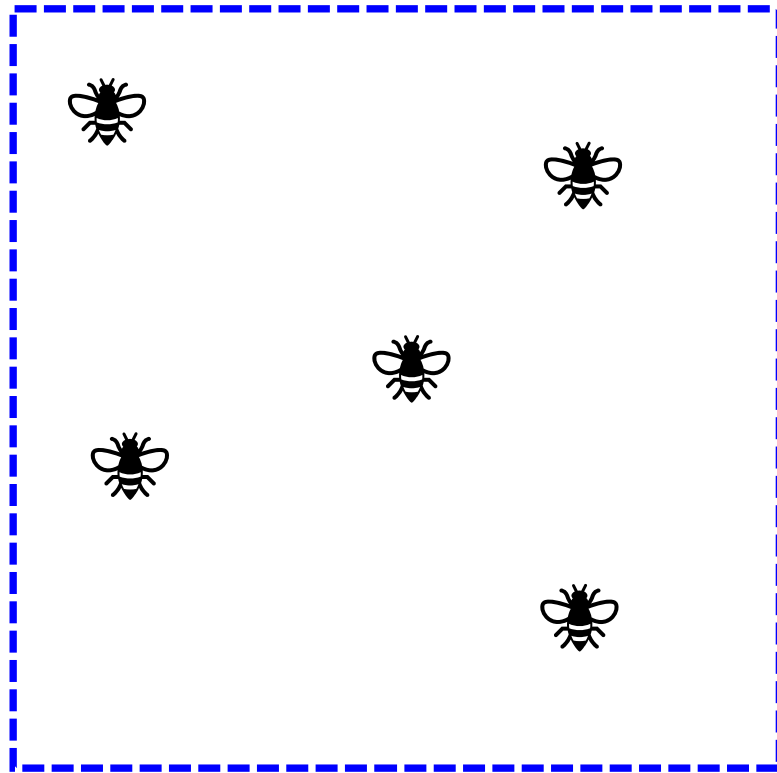
Different factors operating species distribution at different scales !!!



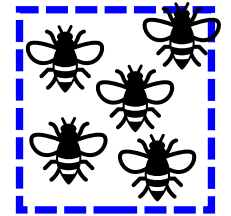
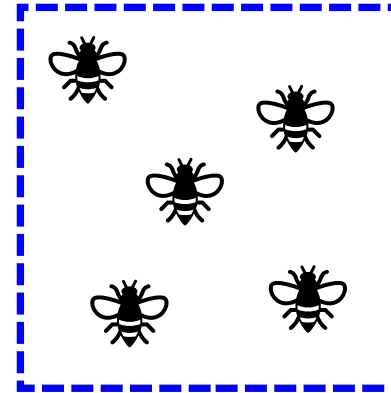
Schematic example of how different factors may affect the distribution of species across varying spatial scales.

Pearson & Dawson 2003

Scale effect



Bioclimate envelope



Climate factor

Non-climate factor
Biotic interaction

Types of factors shaping species distributions

Understanding (insect) species distributions across spatial scales

Hortal et al. 2010 *Ecography*

